Meng Xiang Jia

A Sleep Monitor

Based on LEEGNet and Knowledge Distillation

Number of **304,000,000** adults in China are diagnosed with sleep disorders (Chinese Sociological Association, 2023)



50.4% of patients with insomnia have stroke, hypertension, tumor, heart disease, respiratory (Annual Sleep Report of China 2022)

Main Function: Monitor sleep status and analyze the causes of sleep problems.
Technology: LEEGNet technology was used for the first time to provide users with more precise sleep analysis.
Advantages: Technical innovation, strong portability, multiple application scenarios.....

Price : ***350 RMB***

Market Analysis

Market Potential

Product Introduction

Business Model



Limitations in Traditional Sleeping Monitoring Devices

Sleeping Problems Medical Resource Disparity

Health Risks of Sleep Deprivation

Unaffordable Treatment Cost

Clinical Diagnosis Limitation

B2B Channel	Sales Form
Hospital and other medical institution	Sleep monitoring device & Website
Scientific research institution	Algorithms and collected data

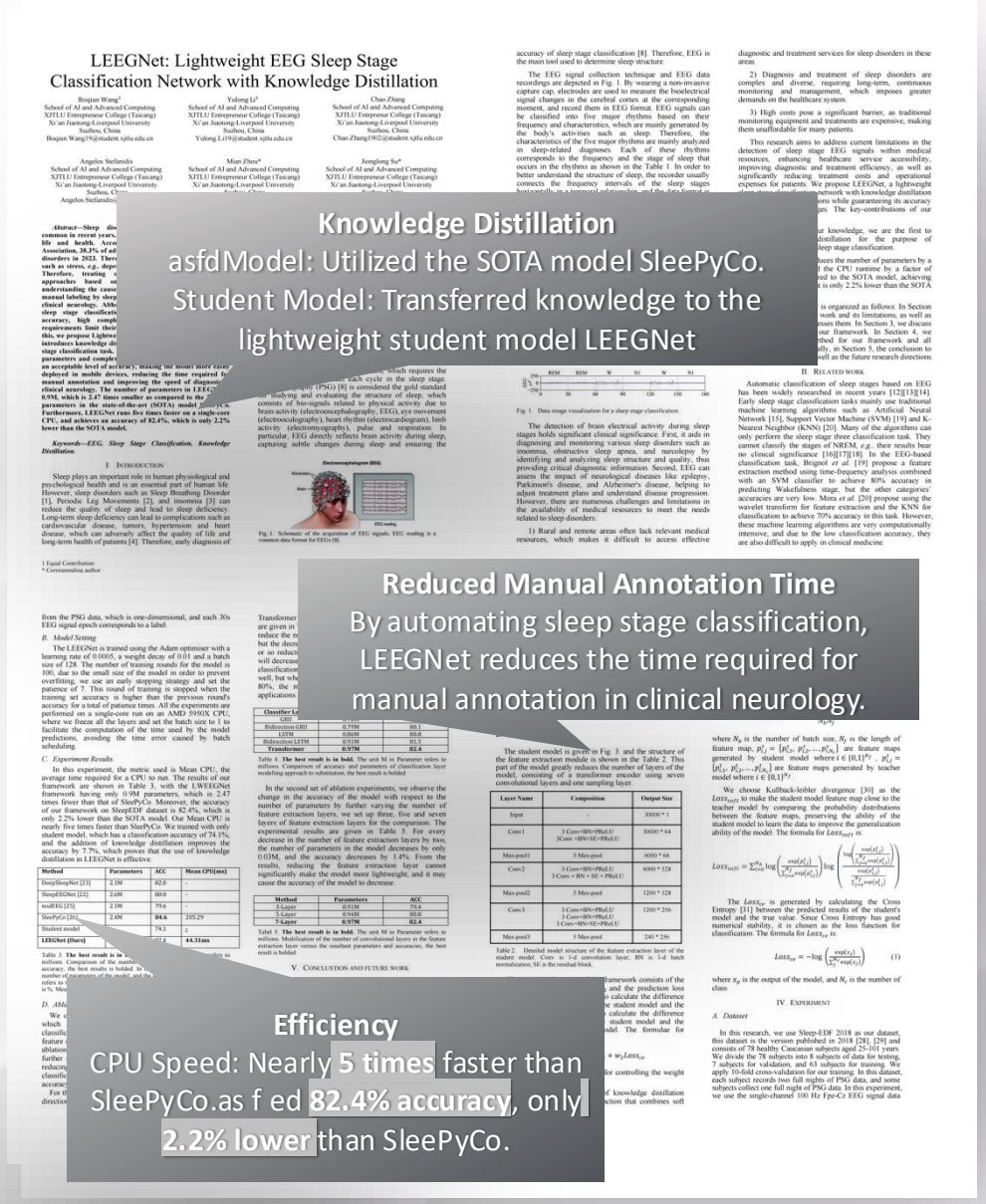
B2C Channel	Sales Form
People with sleeping problems	Sleep monitoring device & APP

B2G Channel	Sales Form
Local governments in rural & remote areas, NGOs	Sleep monitoring device & Website

School: Xi'an Jiaotong-Liverpool University
Members: Shiqi Huang, Wenrui Wu, Zijian He, Chang Liu, Maoshuo Xu, Zihan Zhu

TECHNOLOGY INNOVATION

Innovative points of the thesis



LEEGNet: Lightweight EEG Sleep Stage Classification Network with Knowledge Distillation

Knowledge Distillation

asfdModel: Utilized the SOTA model SleePyCo. Student Model: Transferred knowledge to the lightweight student model LEEGNet

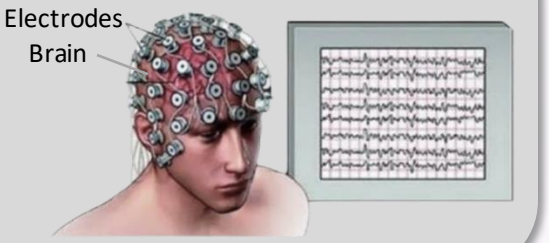
Reduced Manual Annotation Time

By automating sleep stage classification, LEEGNet reduces the time required for manual annotation in clinical neurology.

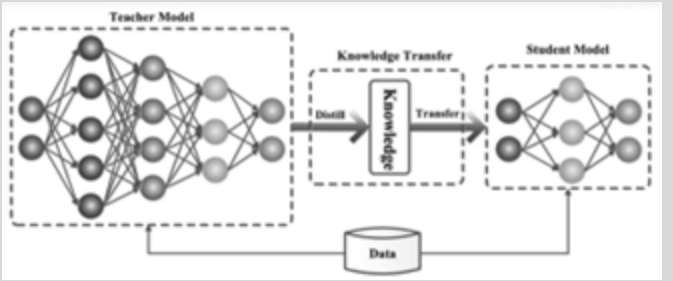
Efficiency

CPU Speed: Nearly **5 times** faster than SleePyCo.as f ed **82.4% accuracy**, only **2.2% lower** than SleePyCo.

Electroencephalogram(EEG)



Knowledge Distillation Model



Technical application process

User wear the sleep monitoring device and fall asleep

→ Brainwave signal collection (EEG)

→ Data transfer to integrated platform (APP/Website)

↓

Apply LEEGNet algorithm and knowledge distillation to processes data

→ Calculate and classify sleep stages to predict the cause of sleep problems

→ Generate a sleep test report, involving solutions

SOCIAL IMPACTS & BENEFITS



Sleep problems need to be solved with cost-effective monitoring devices.

Breaking the technical barriers of traditional sleep detection devices, and creating significant social and clinical value through LEEGNet and knowledge distillation.



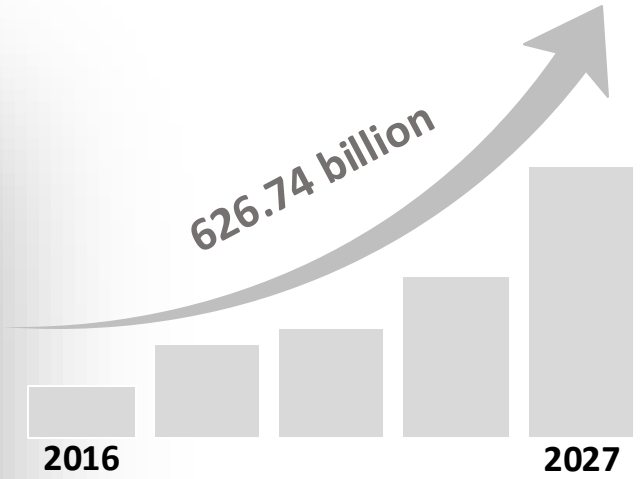
Social Value

1. For Public Welfare: Corporate with local governments in remote and rural areas, providing diagnosis and treatment services.

2. General Public: Call attention to the importance of treatment to sleeping problems.

3. For Scientific Research: This valuable resource is made available to scientific research institutions to further sleep-related research and product innovation.

Expected Market Size



Based on the current situation of sleep disorders in society, our products can solve part of the problem and gain the trust of many people.



(iiMedia Research, 2024)

TEAM COMPOSITION



苏炯龙 Jionglong Su
Deputy Dean at XJTLU AIAC



党康 Kang Dang
Associate Professor of XJTLU AIAC



李冲 Chong Li
Associate Professor of XJTLU AIAC



张博 Bo Zhang
Assistant Professor of XJTLU EEH



Chang Liu

- XJTLU BEng Data Science and Big Data Technology
- Full scholarship in Year 1 Mainly responsible for data processing and analysis



Shiqi Huang

- XJTLU BSc Supply Chain Management
- Mainly responsible for overall organization
- Member of Hear Bridge Technology in XJTLU X3 co-venture



Wenrui Wu

- XJTLU BSc Supply Chain Management
- Served as the management of the community and successfully organized several large-scale activities



Maoshuo Xu

- XJTLU BEng Electronic and electric engineering
- Project assistant intern in Suzhou Jichengsihan Information Technology Company



Zihan Zhu

- XJTLU BEng Data Science and Big Data Technology
- Design the overall video framework and video editing
- Head of ESG Rating Group for Extracurricular Tutor Research Project



Zijian He

- XJTLU BSc Supply Chain Management
- 2 years of experience marketing and management operations
- Founded an education project, radiated more than 40 users

STRATEGIC PLANNING

The level of public acceptance for innovative medical products is suboptimal.

Promote LEEGNet and knowledge distillation technology to the public through product promotion and public welfare activities.

Technical Integration Challenges in Device Deployment

Build an interdisciplinary team consisting of engineers, medical experts, and IT specialists to address Integration of Product and Knowledge issues

Continuous Investment in R&D for Innovation

Raise funds through government subsidies, venture capital, and other means.Continuously optimizing products, improving technology, and enhancing MengXiangJia market competitiveness

